$$\text{Median} = \frac{7+1}{2} = 4^{\text{th}} \text{ positional value}$$

$$\text{Median} = 40 \text{ rooms}$$

Example 5.15

The export of agricultural product in million dollars from a country during eight quarters in 1974 and 1975 was recorded as 29.7, 16.6, 2.3, 14.1, 36.6, 18.7, 3.5, 21.3

Find the median of the given set of values

Solution:

We arrange the data in descending order

36.6, 29.7, 21.3, 18.7, 16.6, 14.1, 3.5, 2.3

$$n = 8 \text{ (even)}$$

$$\text{Median} = \frac{4^{\text{th}}\text{item} + 5^{\text{th}}\text{ item}}{2}$$

$$= \frac{18.7 + 16.6}{2}$$

$$= 17.65 \text{ million dollars}$$

Cumulative Frequency

In a grouped distribution, values are associated with frequencies. The cumulative frequencies are calculated to know the total number of items above or below a certain limit.This is obtained by adding the frequencies successively up to the required level. This cumulative frequencies are useful to calculate median, quartiles, deciles and percentiles.

(b) Median for Discrete grouped data

We can find median using following steps

(i) Calculate the cumulative frequencies

(ii) Find $\frac{N+1}{2}$, Where $N = \sum f$ = total frequencies

(iii) Identify the cumulative frequency just greater than $\frac{N+1}{2}$

(iv) The value of x corresponding to that cumulative frequency $\frac{N+1}{2}$ is the median.

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